

Practical Question: Primary Partition

Question

You can use any available disk (like /dev/sda, /dev/sdb) instead of /dev/nvme0n1, depending on your system's device naming.

You are given a disk /dev/nvme0n1 with free space available.

Perform the following tasks:

Create a primary partition of size 2GB

Format it using the XFS filesystem

Mount it temporarily on the directory /data1

Verify that the partition is mounted

Make the mount permanent using /etc/fstab

Then delete the partition completely, ensuring:

It is unmounted

Removed from /etc/fstab

Deleted from the partition table

Changes applied without reboot

-----Commands-----

```
lsblk
```

```
fdisk /dev/sda → n p 1 +2G w
```

```
partprobe
```

```
mkfs.xfs /dev/sda1
```

```
mkdir /data1
```

```
mount /dev/sda1 /data1
```

```
blkid → add to /etc/fstab
```

```
mount -a
```

```
umount /data1
```

```
remove fstab entry
```

```
fdisk /dev/sda → d 1 w
```

```
partprobe
```

```
lsblk
```

```

20:36:33 | root@localhost | ~ | cmd:116 | cpu: 0.13 | mem:26% | disk:21% | ip
| git: | ssh: | time:0s
🚀 lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda             8:0    0   10G  0 disk
sdb             8:16    0   25G  0 disk
├─sdb1          8:17    0    1M  0 part
├─sdb2          8:18    0    2G  0 part /boot
└─sdb3          8:19    0   23G  0 part
   ├─cs-root    253:0    0 20.5G  0 lvm  /
   └─cs-swap    253:1    0  2.5G  0 lvm  [SWAP]
sr0            11:0    1 1024M  0 rom

```

Perform the following tasks:

Create a primary partition of size 2GB

```

20:37:56 | root@localhost | ~ | cmd:117 | cpu: 0.11 | mem:26% | disk:21% | ip: | git: | ssh: | time:0s
🚀 fdisk /dev/sda

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x17c80010.

Command (m for help): p
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-20971519, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20971519, default 20971519): +2G

Created a new partition 1 of type 'Linux' and of size 2 GiB.

```

```

Command (m for help): p
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sda1		2048	4196351	4194304	2G	83	Linux

```

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

```

```

20:39:42 | root@localhost | ~ | cmd:118 | cpu: 0.01 | mem:26% | disk:21% | ip: | git
: | ssh: | time:0s
🚀 partprobe

```

Format it using XFS filesystem

```

20:40:54 | root@localhost | ~ | cmd:120 | cpu: 0.00 | mem:27% | disk:21% | ip: | g
: | ssh: | time:0s
🚀 mkdir /data1

```

```

20:42:02 | root@localhost | ~ | cmd:121 | cpu: 0.18 | mem:27% | disk:21% | ip: | git: | ssh: | time:0s
🚀 mkfs.xfs /dev/sda1
meta-data=/dev/sda1            isize=512    agcount=4, agsize=131072 blks
                               sectsz=512   attr=2, projid32bit=1
                               crc=1        finobt=1, sparse=1, rmapbt=1
                               =          reflink=1  bigtime=1 inobtcount=1 nnext64=1
                               =          exchange=0 metadir=0
data      =                    bsize=4096   blocks=524288, imaxpct=25
                               =          sunit=0   swidth=0 blks
naming    =version 2          bsize=4096   ascii-ci=0, ftype=1, parent=0
log       =internal log      bsize=4096   blocks=16384, version=2
                               =          sectsz=512  sunit=0 blks, lazy-count=1
realtime  =none              extsz=4096   blocks=0, rtextents=0
                               =          rgcount=0   rgsz=0 extents
                               =          zoned=0     start=0 reserved=0

20:42:35 | root@localhost | ~ | cmd:122 | cpu: 0.11 | mem:27% | disk:21% | ip: | git: | ssh: | time:0s
🚀 lsblk -f
NAME FSTYPE FSVER LABEL UUID                               FSAVAIL FSUSE% MOUNTPOINTS
sda
└─sda1
   xfs                0cfe4039-7f43-475b-816f-e57aa89f1152

```

Mount it temporarily on directory /data1

```
20:43:02 | root@localhost | ~ | cmd:124 | cpu: 0.04 | mem:27% | disk:21% | ip: | git: | ssh: | time:0s
🚀 mount /dev/sda1 /data1
```

Verify that the partition is mounted

```
20:44:14 | root@localhost | ~ | cmd:126 | cpu: 0.04 | mem:27% | disk:21% | ip: | git: | ssh: | time:0s
🚀 lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                  8:0    0   10G  0 disk
└─sda1               8:1    0    2G  0 part /data1
sdb                  8:16    0   25G  0 disk
├─sdb1               8:17    0    1M  0 part
├─sdb2               8:18    0    2G  0 part /boot
└─sdb3               8:19    0   23G  0 part
    └─cs-root        253:0    0 20.5G  0 lvm  /
        └─cs-swap    253:1    0  2.5G  0 lvm  [SWAP]
sr0                  11:0    1 1024M  0 rom
```

(To delete prttn from this point forward use following commands-

umount /data1

remove fstab entry

fdisk /dev/sda → d 1 w

partprobe

lsblk)

Make the mount permanent using /etc/fstab

```
21:02:16 | root@localhost | ~ | cmd:129 | cpu: 0.00 | mem:26% | disk:21% | ip: | git: | ssh: | time:0s
🚀 vim /etc/fstab
```

```
# /etc/fstab
# Created by anaconda on Sun Mar 15 19:41:01 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=e79aa976-2f04-4e51-8942-08170334e649 / xfs defaults 0 0
UUID=b41c8b19-dfe8-4088-8806-ec8fa03b629a /boot xfs defaults 0 0
UUID=de55cc71-342c-4646-817d-ae6a99e96439 none swap defaults 0 0
#-----

#/dev/sda1: UUID="a1b98820-f4c5-43ad-ab2c-b5585b83d06f" BLOCK_SIZE="512" TYPE="xfs" PARTUUID="17c80010-01"
#Safe practice is to use UUID
#UUID=a1b98820-f4c5-43ad-ab2c-b5585b83d06f /data1 xfs defaults 0 0

/dev/sda1 /data1 xfs defaults 0 0
```

```
21:07:25 | root@localhost | ~ | cmd:131 | cpu: 0.02 | mem:28% | disk:21% |
| git: | ssh: | time:0s
🚀 systemctl daemon-reload
```

```
21:08:45 | root@localhost | ~ | cmd:133 | cpu: 0.13 | mem:28%
| git: | ssh: | time:0s
🚀 mount -a
```

Then delete the partition completely, ensuring:

It is unmounted

```
21:11:45 | root@localhost | ~ | cmd:136 | cpu: 0.02 | mem:28% |
time:0s
🚀 umount /dev/sda1
```

Removed from /etc/fstab

[illegible]

🚀 `systemctl daemon-reload`

```
21:13:04 | root@localhost | ~ | cmd:139 | cpu: 0.00 | mem:28% | disk:21% | ip: | git: |  
time:0s
```

🚀 df -h

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/mapper/cs-root	21G	4.3G	17G	21%	/
devtmpfs	2.8G	0	2.8G	0%	/dev
tmpfs	2.8G	84K	2.8G	1%	/dev/shm
tmpfs	1.2G	9.5M	1.2G	1%	/run
tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
/dev/sdb2	2.0G	446M	1.6G	23%	/boot
tmpfs	570M	116K	570M	1%	/run/user/0

Deleted from the partition table

```
21:14:22 | root@localhost | ~ | cmd:140 | cpu: 0.08 | mem:28% | disk:21%  
🚀 fdisk /dev/sda  
  
Welcome to fdisk (util-linux 2.40.2).  
Changes will remain in memory only, until you decide to write them.  
Be careful before using the write command.  
  
Command (m for help): p  
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors  
Disk model: VMware Virtual S  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes  
Disklabel type: dos  
Disk identifier: 0x17c80010  
  


| Device    | Boot | Start | End     | Sectors | Size | Id | Type  |
|-----------|------|-------|---------|---------|------|----|-------|
| /dev/sda1 |      | 2048  | 4196351 | 4194304 | 2G   | 83 | Linux |

  
Command (m for help): d  
Selected partition 1  
Partition 1 has been deleted.
```

```
Command (m for help): p  
  
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors  
Disk model: VMware Virtual S  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes  
Disklabel type: dos  
Disk identifier: 0x17c80010  
  
Command (m for help): w  
  
The partition table has been altered.  
Calling ioctl() to re-read partition table.  
Syncing disks.
```

Changes applied without reboot

```
21:16:39 | root@localhost | ~ | cmd:141 |  
🚀 partprobe
```

```
21:16:46 | root@localhost | ~ | cmd:142 | cpu: 0.01 | mem:28% |  
🚀 lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	10G	0	disk	
sdb	8:16	0	25G	0	disk	
├─sdb1	8:17	0	1M	0	part	
├─sdb2	8:18	0	2G	0	part	/boot
└─sdb3	8:19	0	23G	0	part	
├─cs-root	253:0	0	20.5G	0	lvm	/
└─cs-swap	253:1	0	2.5G	0	lvm	[SWAP]
sr0	11:0	1	1024M	0	rom	